Nand2Tetris Project 3

HDL Code + Chip Simulation

Component	HDL Code	Screenshot
Bit	Included	Included
Register	Included	Included
RAM8	Included	Included
RAM64	Included	Included
RAM512	Included	Included
RAM4K	Included	Included
RAM16K	Included	Included
PC	Included	Included

By: Himanchal Raghuvanshi

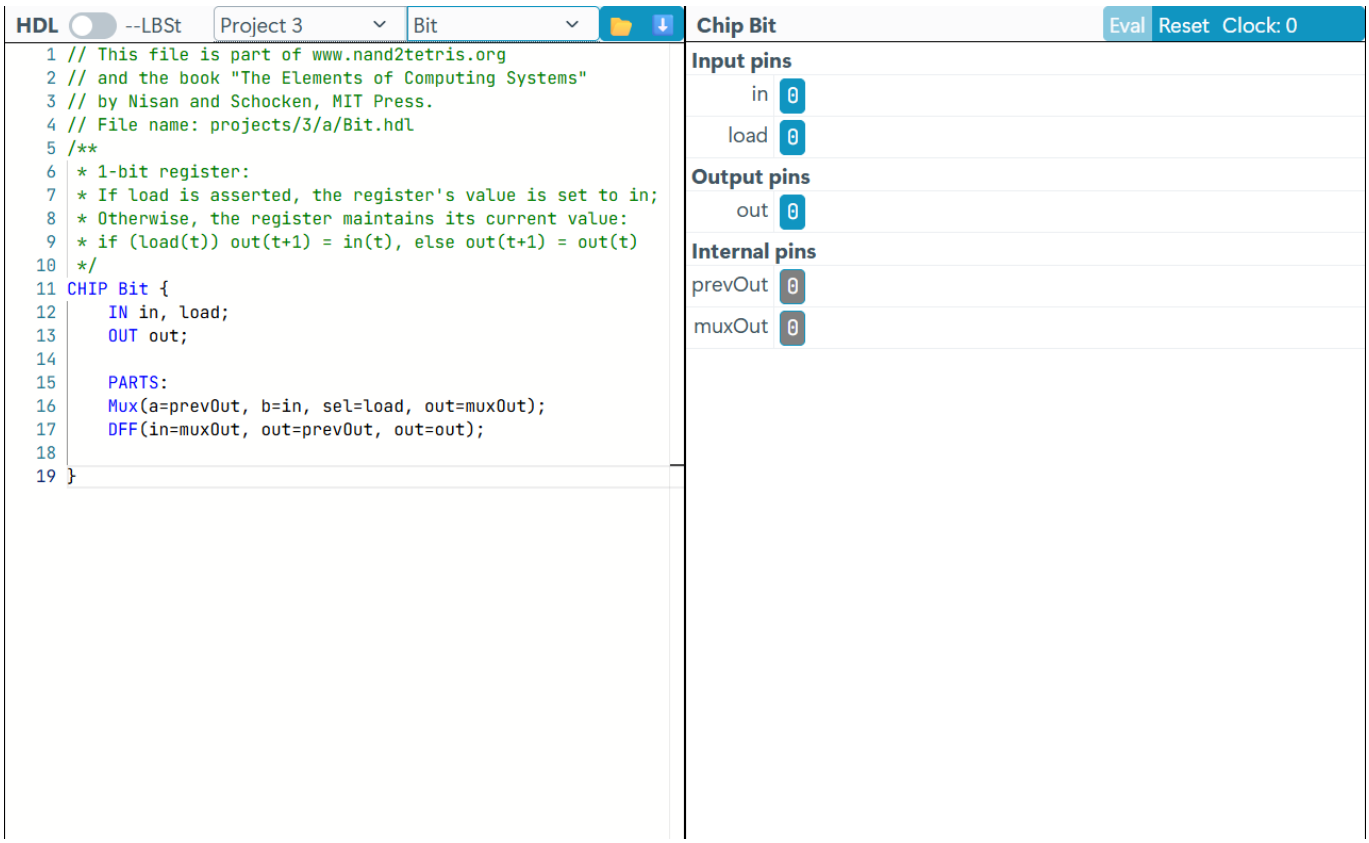
1. Bit.hdl

HDL Source Code

```
01 // This file is part of www.nand2tetris.org
02 // and the book "The Elements of Computing Systems"
03 // by Nisan and Schocken, MIT Press.
04 // File name: projects/3/a/Bit.hdl
05 /**
06 * 1-bit register:
07 * If load is asserted, the register's value is set to in;
08 * Otherwise, the register maintains its current value:
09 * if (load(t)) out(t+1) = in(t), else out(t+1) = out(t)
10 */
11 CHIP Bit {
12 IN in, load;
13 OUT out;
14
15 PARTS:
16 Mux(a=prevOut, b=in, sel=load, out=muxOut);
17 DFF(in=muxOut, out=prevOut, out=out);
18
19 }
```

File: Bit(1).hdl | Chip: Bit

1. Bit — Simulation Screenshot



Provided screenshot from the HDL/chip simulator.

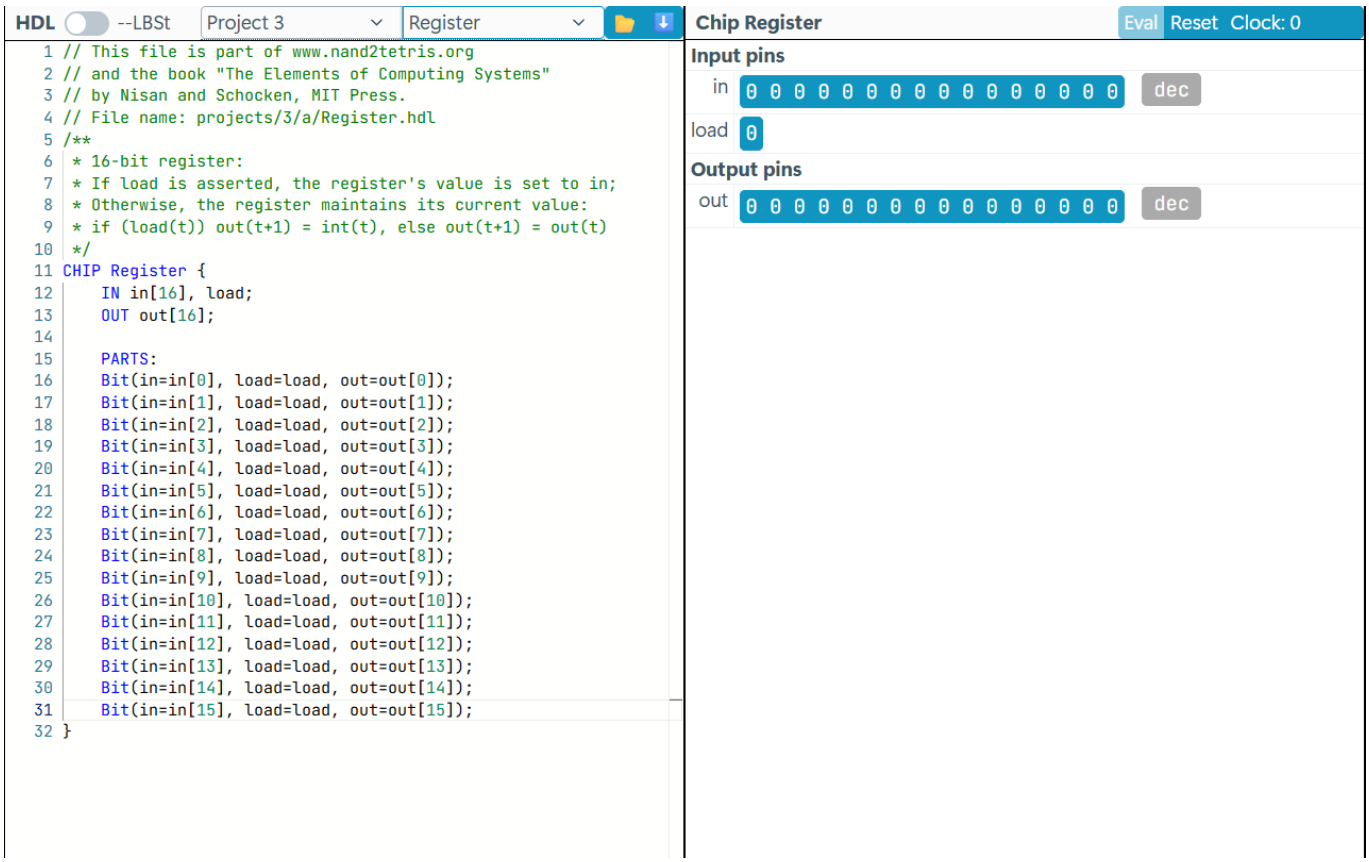
2. Register.hdl

HDL Source Code

```
01 // This file is part of www.nand2tetris.org
02 // and the book "The Elements of Computing Systems"
03 // by Nisan and Schocken, MIT Press.
04 // File name: projects/3/a/Register.hdl
05 /**
06 * 16-bit register:
07 * If load is asserted, the register's value is set to in;
08 * Otherwise, the register maintains its current value:
09 * if (load(t)) out(t+1) = int(t), else out(t+1) = out(t)
10 */
11 CHIP Register {
12 IN in[16], load;
13 OUT out[16];
14
15 PARTS:
16 Bit(in=in[0], load=load, out=out[0]);
17 Bit(in=in[1], load=load, out=out[1]);
18 Bit(in=in[2], load=load, out=out[2]);
19 Bit(in=in[3], load=load, out=out[3]);
20 Bit(in=in[4], load=load, out=out[4]);
21 Bit(in=in[5], load=load, out=out[5]);
22 Bit(in=in[6], load=load, out=out[6]);
23 Bit(in=in[7], load=load, out=out[7]);
24 Bit(in=in[8], load=load, out=out[8]);
25 Bit(in=in[9], load=load, out=out[9]);
26 Bit(in=in[10], load=load, out=out[10]);
27 Bit(in=in[11], load=load, out=out[11]);
28 Bit(in=in[12], load=load, out=out[12]);
29 Bit(in=in[13], load=load, out=out[13]);
30 Bit(in=in[14], load=load, out=out[14]);
31 Bit(in=in[15], load=load, out=out[15]);
32 }
```

File: Register(1).hdl | Chip: Register

2. Register — Simulation Screenshot



Provided screenshot from the HDL/chip simulator.

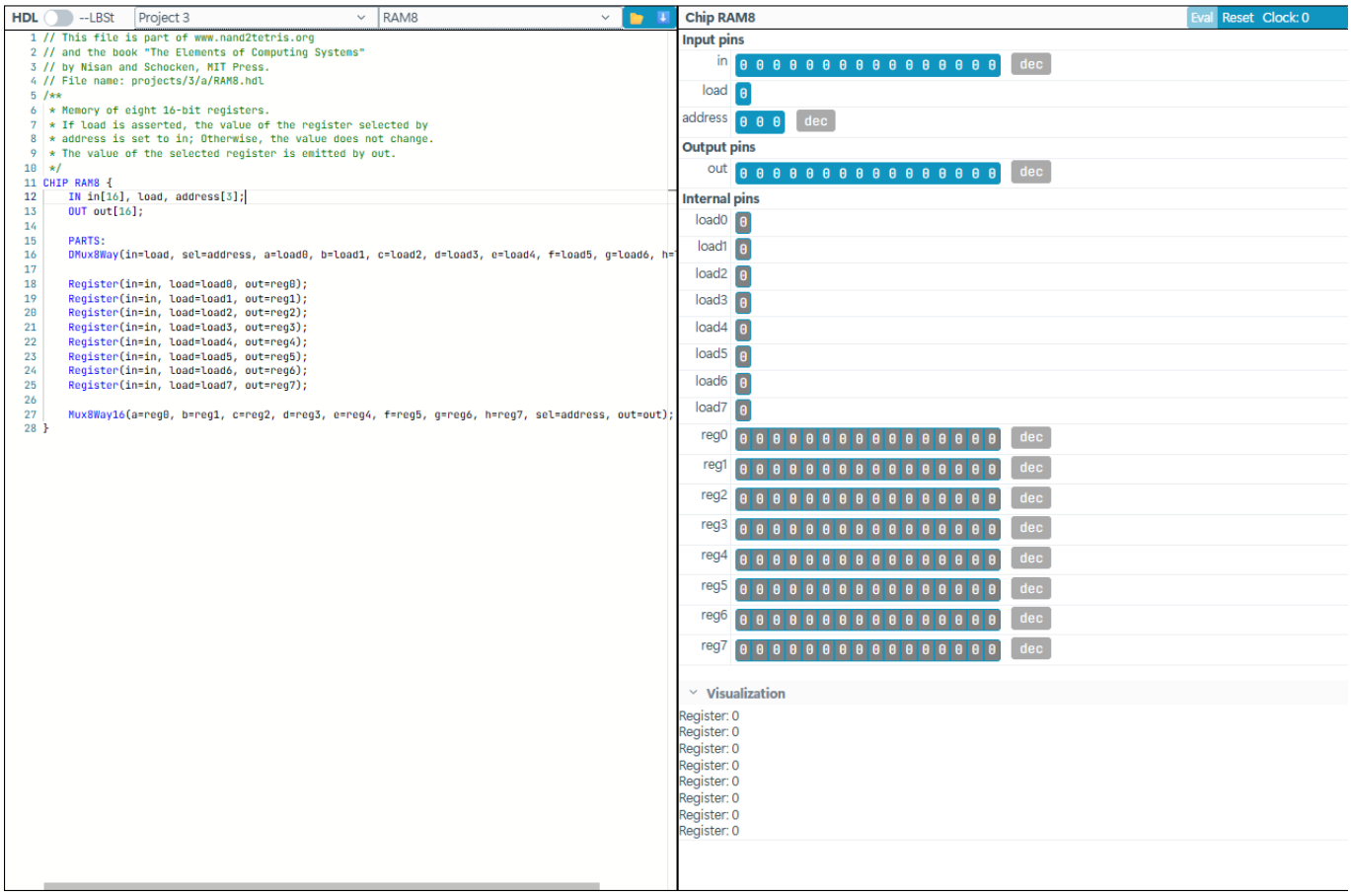
3. RAM8.hdl

HDL Source Code

```
01 // This file is part of www.nand2tetris.org
02 // and the book "The Elements of Computing Systems"
03 // by Nisan and Schocken, MIT Press.
04 // File name: projects/3/a/RAM8.hdl
05 /**
06 * Memory of eight 16-bit registers.
07 * If load is asserted, the value of the register selected by
08 * address is set to in; Otherwise, the value does not change.
09 * The value of the selected register is emitted by out.
10 */
11 CHIP RAM8 {
12 IN in[16], load, address[3];
13 OUT out[16];
14
15 PARTS:
16 DMux8Way(in=load, sel=address, a=load0, b=load1, c=load2, d=load3, e=load4, f=load5, g=load6,
h=load7);
17
18 Register(in=in, load=load0, out=reg0);
19 Register(in=in, load=load1, out=reg1);
20 Register(in=in, load=load2, out=reg2);
21 Register(in=in, load=load3, out=reg3);
22 Register(in=in, load=load4, out=reg4);
23 Register(in=in, load=load5, out=reg5);
24 Register(in=in, load=load6, out=reg6);
25 Register(in=in, load=load7, out=reg7);
26
27 Mux8Way16(a=reg0, b=reg1, c=reg2, d=reg3, e=reg4, f=reg5, g=reg6, h=reg7, sel=address, out=out);
28 }
```

File: RAM8(1).hdl | Chip: RAM8

3. RAM8 — Simulation Screenshot



Provided screenshot from the HDL/chip simulator.

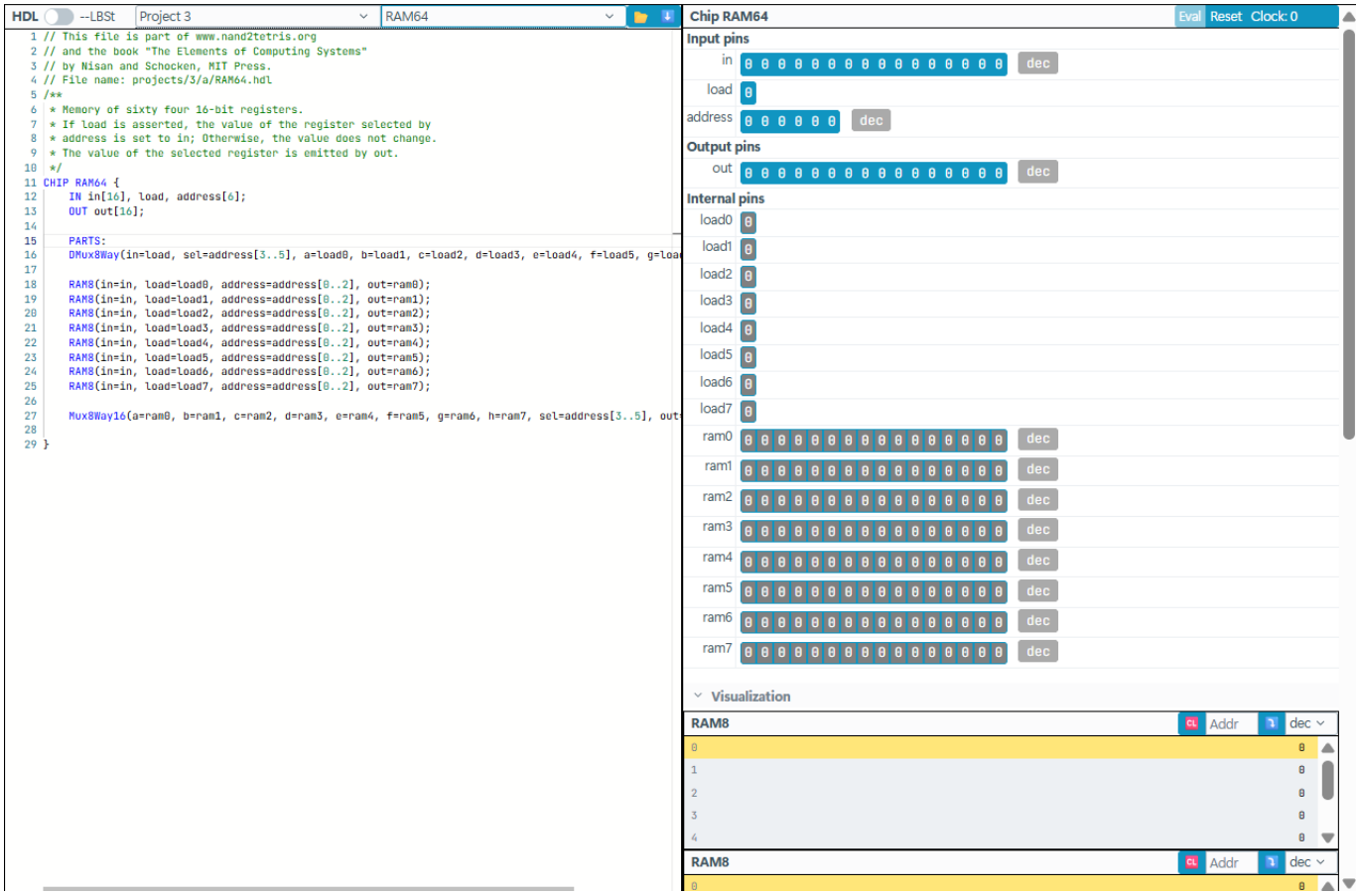
4. RAM64.hdl

HDL Source Code

```
01 // This file is part of www.nand2tetris.org
02 // and the book "The Elements of Computing Systems"
03 // by Nisan and Schocken, MIT Press.
04 // File name: projects/3/a/RAM64.hdl
05 /**
06 * Memory of sixty four 16-bit registers.
07 * If load is asserted, the value of the register selected by
08 * address is set to in; Otherwise, the value does not change.
09 * The value of the selected register is emitted by out.
10 */
11 CHIP RAM64 {
12 IN in[16], load, address[6];
13 OUT out[16];
14
15 PARTS:
16 DMux8Way(in=load, sel=address[3..5], a=load0, b=load1, c=load2, d=load3, e=load4, f=load5, g=load6,
h=load7);
17
18 RAM8(in=in, load=load0, address=address[0..2], out=ram0);
19 RAM8(in=in, load=load1, address=address[0..2], out=ram1);
20 RAM8(in=in, load=load2, address=address[0..2], out=ram2);
21 RAM8(in=in, load=load3, address=address[0..2], out=ram3);
22 RAM8(in=in, load=load4, address=address[0..2], out=ram4);
23 RAM8(in=in, load=load5, address=address[0..2], out=ram5);
24 RAM8(in=in, load=load6, address=address[0..2], out=ram6);
25 RAM8(in=in, load=load7, address=address[0..2], out=ram7);
26
27 Mux8Way16(a=ram0, b=ram1, c=ram2, d=ram3, e=ram4, f=ram5, g=ram6, h=ram7, sel=address[3..5],
out=out);
28
29 }
```

File: RAM64(1).hdl | Chip: RAM64

4. RAM64 — Simulation Screenshot



Provided screenshot from the HDL/chip simulator.

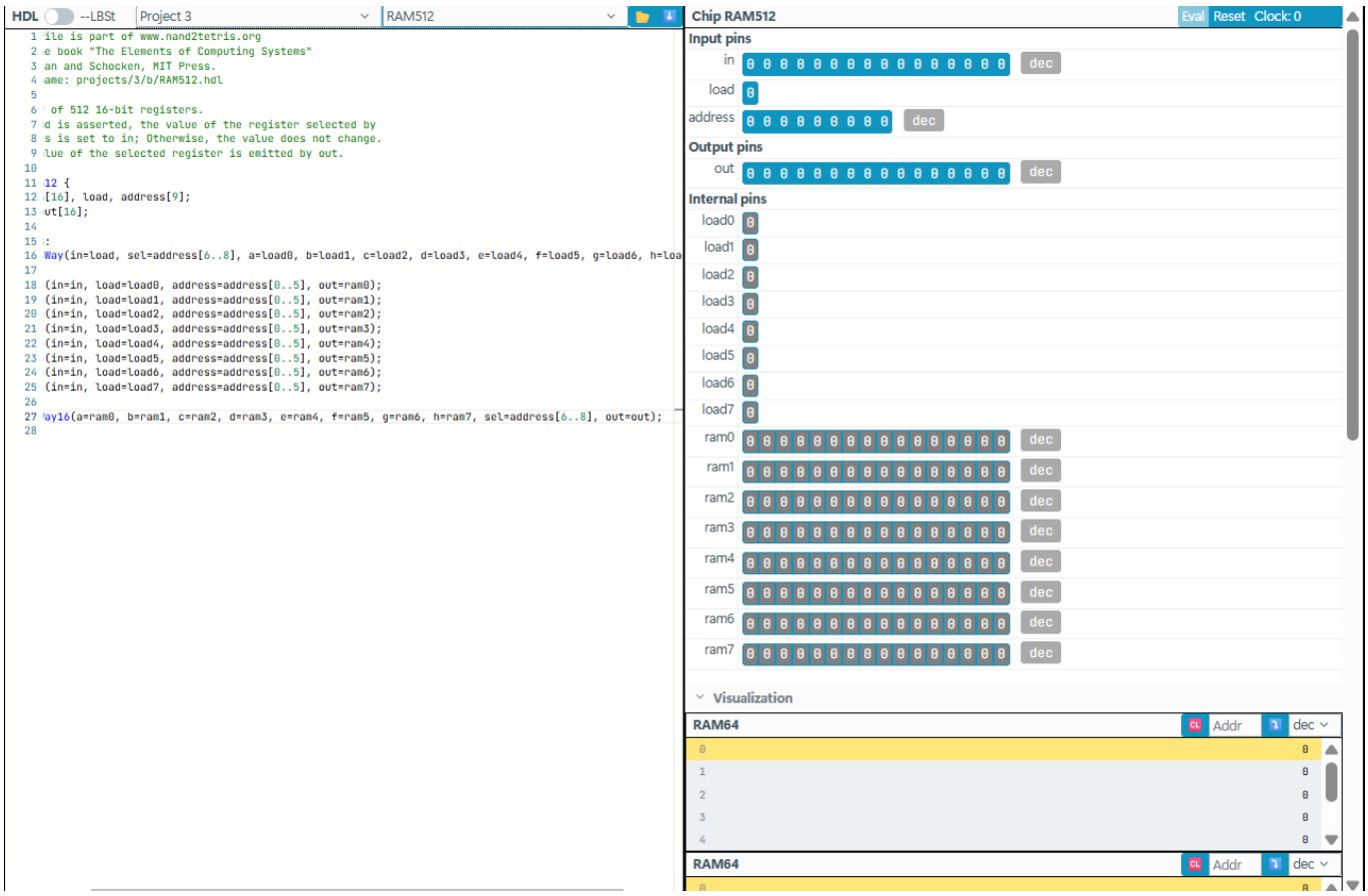
5. RAM512.hdl

HDL Source Code

```
01 // This file is part of www.nand2tetris.org
02 // and the book "The Elements of Computing Systems"
03 // by Nisan and Schocken, MIT Press.
04 // File name: projects/3/b/RAM512.hdl
05 /**
06 * Memory of 512 16-bit registers.
07 * If load is asserted, the value of the register selected by
08 * address is set to in; Otherwise, the value does not change.
09 * The value of the selected register is emitted by out.
10 */
11 CHIP RAM512 {
12 IN in[16], load, address[9];
13 OUT out[16];
14
15 PARTS:
16 DMux8Way(in=load, sel=address[6..8], a=load0, b=load1, c=load2, d=load3, e=load4, f=load5, g=load6,
h=load7);
17
18 RAM64(in=in, load=load0, address=address[0..5], out=ram0);
19 RAM64(in=in, load=load1, address=address[0..5], out=ram1);
20 RAM64(in=in, load=load2, address=address[0..5], out=ram2);
21 RAM64(in=in, load=load3, address=address[0..5], out=ram3);
22 RAM64(in=in, load=load4, address=address[0..5], out=ram4);
23 RAM64(in=in, load=load5, address=address[0..5], out=ram5);
24 RAM64(in=in, load=load6, address=address[0..5], out=ram6);
25 RAM64(in=in, load=load7, address=address[0..5], out=ram7);
26
27 Mux8Way16(a=ram0, b=ram1, c=ram2, d=ram3, e=ram4, f=ram5, g=ram6, h=ram7, sel=address[6..8],
out=out);
28 }
```

File: RAM512(1).hdl | Chip: RAM512

5. RAM512 — Simulation Screenshot



Provided screenshot from the HDL/chip simulator.

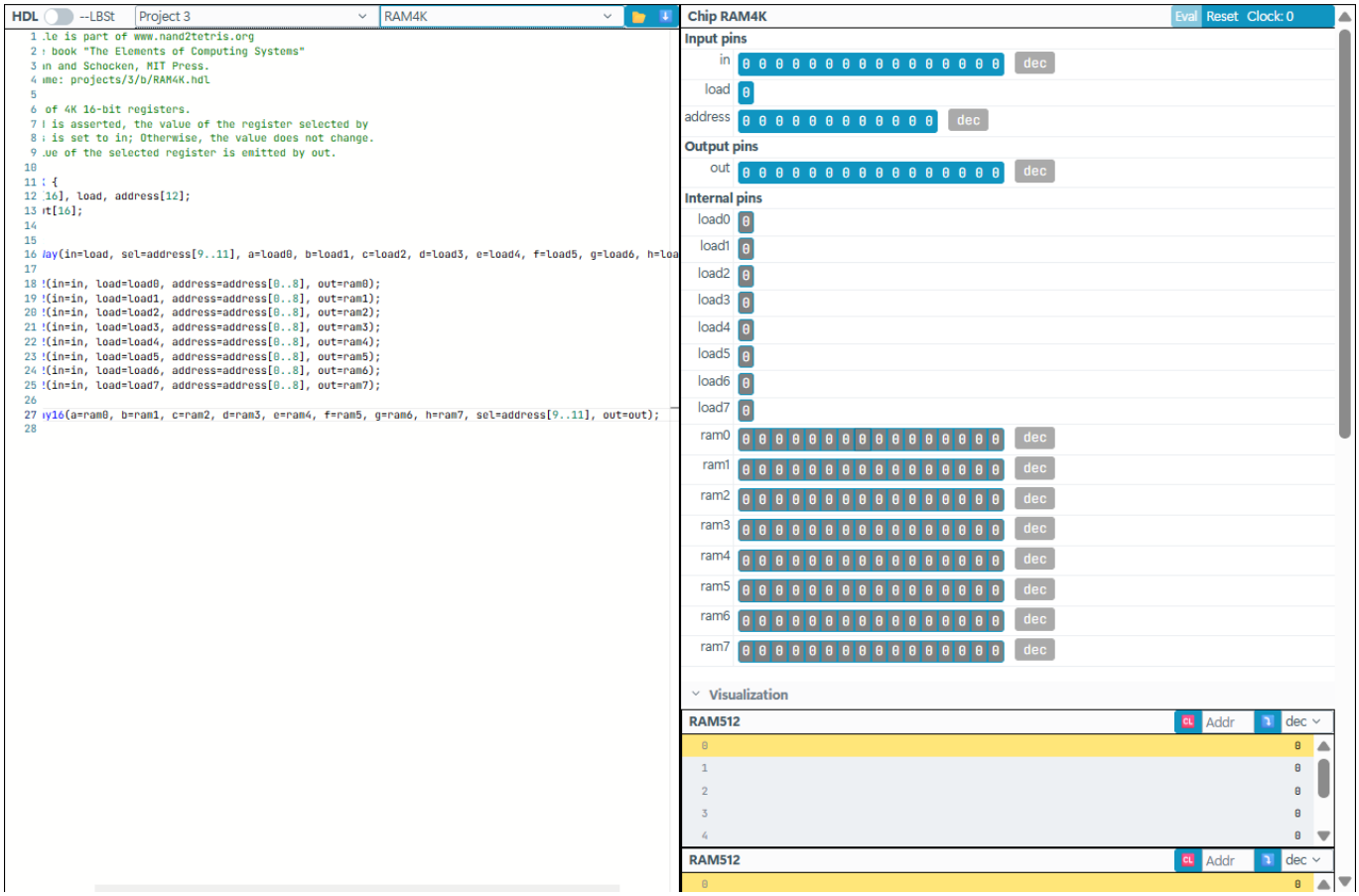
6. RAM4K.hdl

HDL Source Code

```
01 // This file is part of www.nand2tetris.org
02 // and the book "The Elements of Computing Systems"
03 // by Nisan and Schocken, MIT Press.
04 // File name: projects/3/b/RAM4K.hdl
05 /**
06 * Memory of 4K 16-bit registers.
07 * If load is asserted, the value of the register selected by
08 * address is set to in; Otherwise, the value does not change.
09 * The value of the selected register is emitted by out.
10 */
11 CHIP RAM4K {
12 IN in[16], load, address[12];
13 OUT out[16];
14
15 PARTS:
16 DMux8Way(in=load, sel=address[9..11], a=load0, b=load1, c=load2, d=load3, e=load4, f=load5, g=load6,
h=load7);
17
18 RAM512(in=in, load=load0, address=address[0..8], out=ram0);
19 RAM512(in=in, load=load1, address=address[0..8], out=ram1);
20 RAM512(in=in, load=load2, address=address[0..8], out=ram2);
21 RAM512(in=in, load=load3, address=address[0..8], out=ram3);
22 RAM512(in=in, load=load4, address=address[0..8], out=ram4);
23 RAM512(in=in, load=load5, address=address[0..8], out=ram5);
24 RAM512(in=in, load=load6, address=address[0..8], out=ram6);
25 RAM512(in=in, load=load7, address=address[0..8], out=ram7);
26
27 Mux8Way16(a=ram0, b=ram1, c=ram2, d=ram3, e=ram4, f=ram5, g=ram6, h=ram7, sel=address[9..11],
out=out);
28 }
```

File: RAM4K(1).hdl | Chip: RAM4K

6. RAM4K — Simulation Screenshot



Provided screenshot from the HDL/chip simulator.

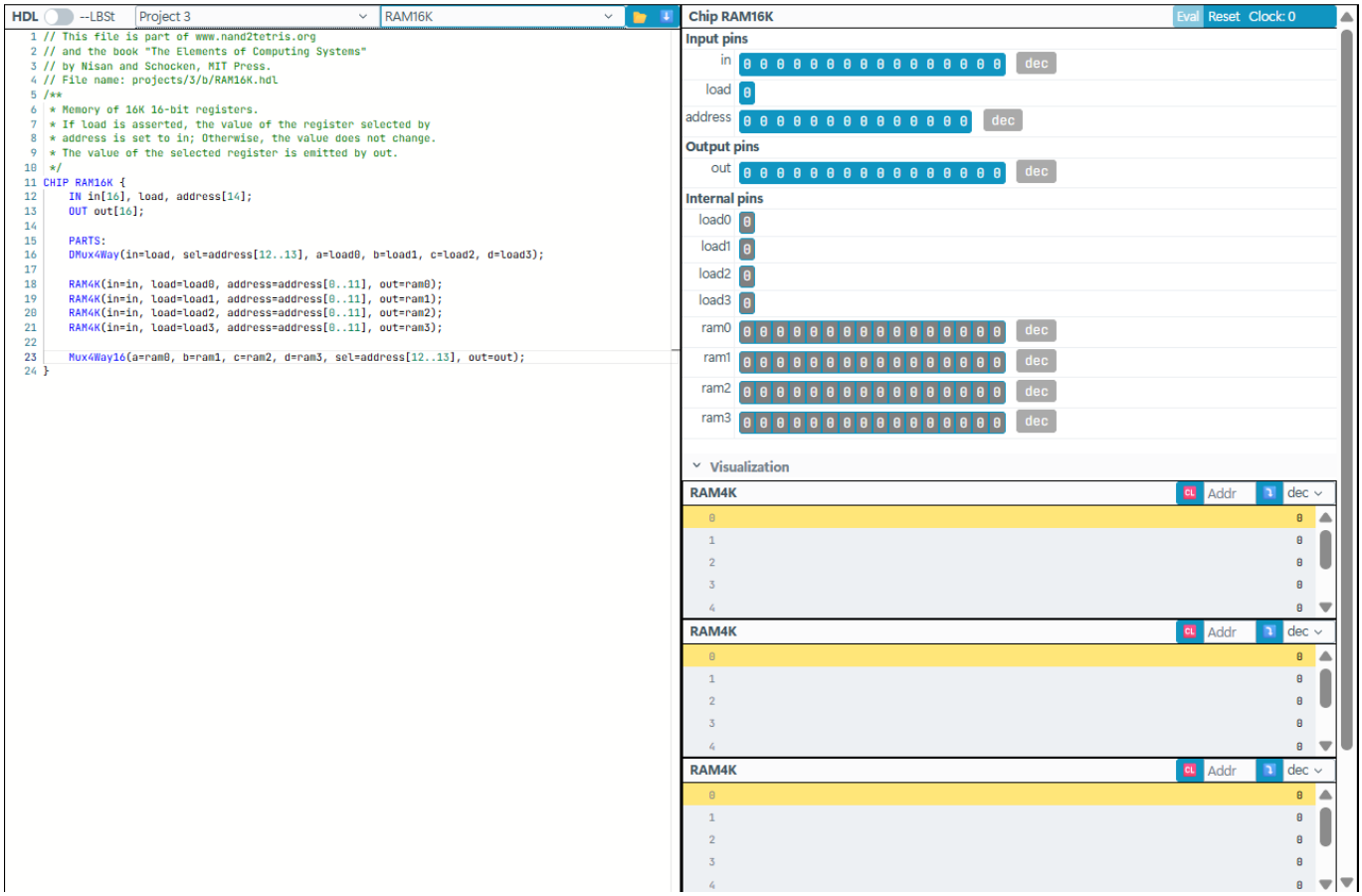
7. RAM16K.hdl

HDL Source Code

```
01 // This file is part of www.nand2tetris.org
02 // and the book "The Elements of Computing Systems"
03 // by Nisan and Schocken, MIT Press.
04 // File name: projects/3/b/RAM16K.hdl
05 /**
06 * Memory of 16K 16-bit registers.
07 * If load is asserted, the value of the register selected by
08 * address is set to in; Otherwise, the value does not change.
09 * The value of the selected register is emitted by out.
10 */
11 CHIP RAM16K {
12 IN in[16], load, address[14];
13 OUT out[16];
14
15 PARTS:
16 DMux4Way(in=load, sel=address[12..13], a=load0, b=load1, c=load2, d=load3);
17
18 RAM4K(in=in, load=load0, address=address[0..11], out=ram0);
19 RAM4K(in=in, load=load1, address=address[0..11], out=ram1);
20 RAM4K(in=in, load=load2, address=address[0..11], out=ram2);
21 RAM4K(in=in, load=load3, address=address[0..11], out=ram3);
22
23 Mux4Way16(a=ram0, b=ram1, c=ram2, d=ram3, sel=address[12..13], out=out);
24 }
```

File: RAM16K(1).hdl | Chip: RAM16K

7. RAM16K — Simulation Screenshot



Provided screenshot from the HDL/chip simulator.

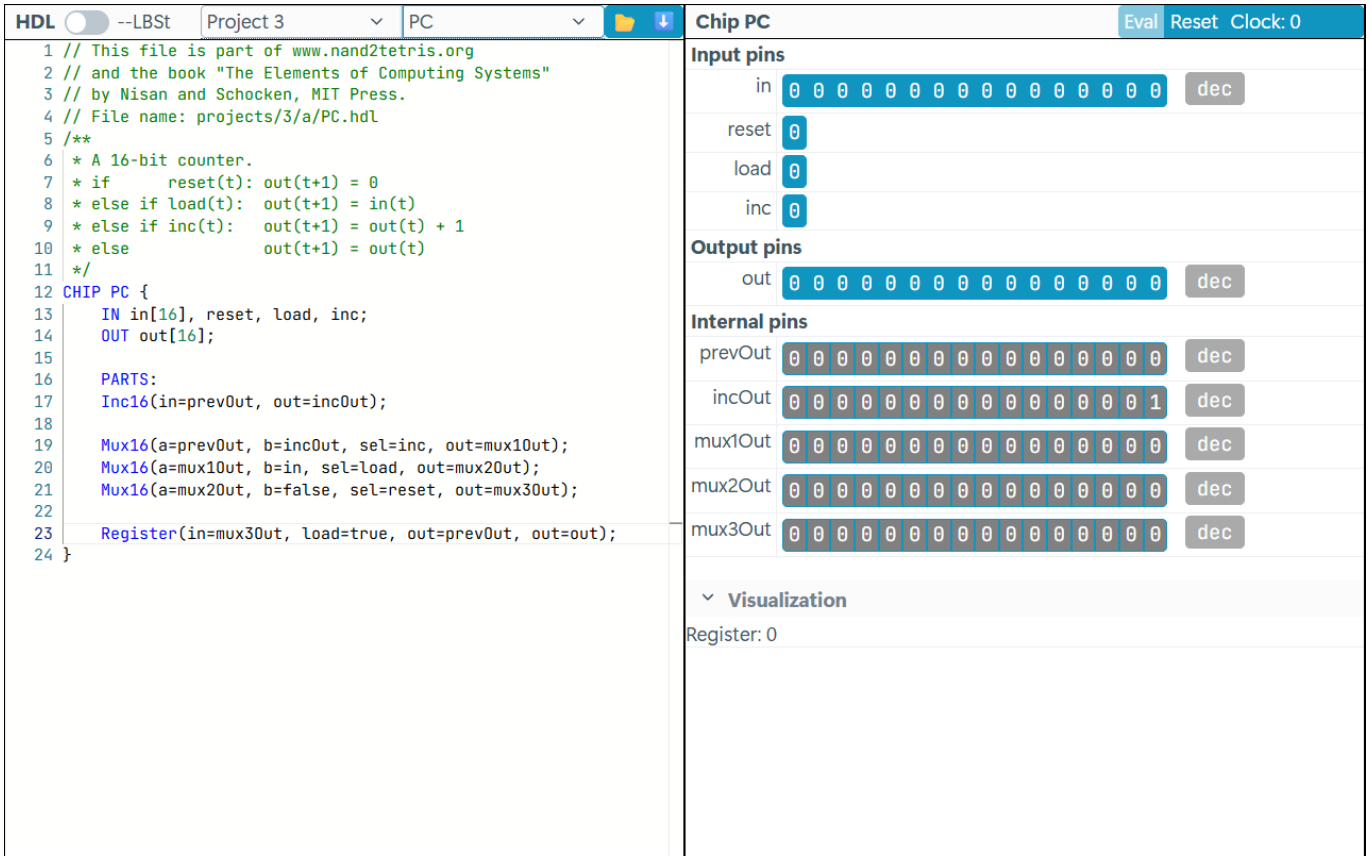
8. PC.hdl

HDL Source Code

```
01 // This file is part of www.nand2tetris.org
02 // and the book "The Elements of Computing Systems"
03 // by Nisan and Schocken, MIT Press.
04 // File name: projects/3/a/PC.hdl
05 /**
06 * A 16-bit counter.
07 * if reset(t): out(t+1) = 0
08 * else if load(t): out(t+1) = in(t)
09 * else if inc(t): out(t+1) = out(t) + 1
10 * else out(t+1) = out(t)
11 */
12 CHIP PC {
13 IN in[16], reset, load, inc;
14 OUT out[16];
15
16 PARTS:
17 Inc16(in=prevOut, out=incOut);
18
19 Mux16(a=prevOut, b=incOut, sel=inc, out=mux1Out);
20 Mux16(a=mux1Out, b=in, sel=load, out=mux2Out);
21 Mux16(a=mux2Out, b=false, sel=reset, out=mux3Out);
22
23 Register(in=mux3Out, load=true, out=prevOut, out=out);
24 }
```

File: PC(1).hdl | Chip: PC

8. PC — Simulation Screenshot



Provided screenshot from the HDL/chip simulator.